

1. Primary traffic data (field-collected)

- **Classified directional volume counts** — 15-minute intervals, both directions, separated by vehicle class (cars, matatus/PSVs, buses, HGVs, motorcycles, NMT). Cover full AM peak (roughly 6:00–9:00) and PM peak (16:00–19:00) at minimum; ideally a full 24-hr count on a representative weekday (Tue–Thu) plus one weekend day for contrast.
- **Turning movement counts** at Muthaiga Roundabout and Garden City Interchange — needed to separate through-traffic (which tidal flow would serve) from turning traffic.
- **Spot speed / travel-time runs** in both directions, peak and off-peak.
- **Per-lane volume distribution** — which lanes carry how much, needed for the V/C and geometric-adequacy checks.

This block feeds DFR and PHF directly — it's the data you can't substitute with anything secondary.

2. Geometric/roadway inventory data

- Current number of lanes per direction and total cross-section
- Lane widths, shoulder presence/width
- Median type and width (raised, paved, flush, existing barrier) — critical, since a raised median without movable barriers changes feasibility a lot
- Horizontal/vertical alignment details relevant to sight distance
- Access-control classification of the segment (fully vs. partially controlled)
- Existing signage, lane markings, and any ITS infrastructure (CCTV, VMS) already in place
- As-built or design drawings for this A2 stretch, plus interchange/ramp geometry at both boundary points

Source: KeNHA (Kenya National Highways Authority) design/as-built drawings — this is the main formal data request you'll need to make.

3. Access-point inventory

- Count and GPS location (chainage) of every driveway, side road, matatu stage, bus stop, fuel station entrance, and commercial access along the segment
- Spacing between consecutive access points
- Location of U-turn bays/median openings

Mostly desk-based via satellite imagery, verified with a site walk.

4. Safety data

- Crash records for the segment, ideally 3+ years, with location, severity, time of day, and contributing factors
- Existing traffic control devices at access points and the roundabout/interchange
- Pedestrian/NMT crossing volumes and locations

- Any official blackspot designation for this stretch

Source: NTSA (National Transport and Safety Authority) and/or Kenya Police traffic records — this is a formal request, and response times can be slow, so start this one early.

5. Secondary/published data

- KeNHA's published traffic census figures or prior ADT data for Thika Superhighway
- Any existing feasibility, capacity, or corridor studies on this route (JICA and World Bank both funded Thika Superhighway work historically — worth checking their public report archives)
- Nairobi Metropolitan transport master plan data relevant to this corridor (NaMATA plans, NIUPLAN) for growth/demand context
- Public transport route data — matatu/bus routes and stage locations along the corridor
- Land-use context around Muthaiga and Garden City Mall (major trip generator right at your segment boundary)

6. Satellite/GIS-derived data

- High-resolution imagery (Google Earth Pro export is usually sufficient) to measure segment length, lane counts, and access-point spacing where field verification isn't practical
- Precise GPS coordinates for segment start/end (Muthaiga Roundabout, Garden City Interchange) and any key reference points

7. Regulatory/institutional data

- Kenya's road geometric design standards (KeNHA design manual) — needed to judge whether your geometric-adequacy scoring aligns with local standards, not just AASHTO/FHWA
- Confirmation of whether tidal flow/lane reversal is addressed anywhere in Kenyan traffic law or KeNHA operational guidance — matters for your "feasibility" framing, since technical feasibility and regulatory feasibility are different questions

8. Validation-support data

- A reference or benchmark case (published tidal-flow feasibility assessment elsewhere, even outside Kenya) to sanity-check that your manual worksheet and software agree with a known-good calculation
- Your supervisor's criteria/checklist for what counts as adequate validation — worth getting this in writing before you build the validation chapter, not after